Certifiably Biased: An In-Depth Analysis of a Common Criteria EAL4+ Certified TRNG

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***Abstract*— This paper reports the first in-depth analysis of the DESFire EV1's EAL4+ certified TRNG and raises some difficult questions regarding the certification of non-deterministic random number generators. We start by analyzing the quality of the purportedly true random number generator (TRNG) on the DESFire EV1 card. Clear and consistent biases are identified, despite good performance in most randomness tests. These statistical tests, commonly used in popular certification processes, such as Common Criteria EAL4+, are found not to be able to detect these anomalies. The means we employ for the detection and characterization of the bias are explored, highlighting both novel and existing ways of spotting deficient TRNG output. Further analysis shows systemic issues affecting TRNG output at the byte level, for which we have developed an accurate explanation. Our results have been acknowledged by the manufacturer, after responsible disclosure.**

***Index Terms*— Random number generation, RFID, security, common criteria, statistical testing.**

I. Introduction

RADIO Frequency Identification (RFID) systems have become embedded in many aspects of our daily lives. For example, the convenience offered by contactless payment cards has made this a desirable technology to both customers and service providers. This is apparent in systems like Transport for London's (TfL) Oyster card. TfL statistics show that in the 2015/16 financial period, more than 8 million Oyster cards were sold [1]. Between 2 and 3 million Oyster card swipes were recorded per day for the same period [2]. All such cards have a cash value associated with them, as fares are prepaid. Any payment system must be secure against a range of threats including tampering, cloning, and traceability attacks. The Mifare Classic failed to defend against these attacks and was publicly and comprehensively dismantled due to a weak cipher (CRYPTO-1) [3]. This NXP produced card relied on security through obscurity, hiding a flawed crypto-system behind a closed-standard. A manipulable Pseudo Random Number Generator (PRNG) further contributed to the demise of these cards [4], [5]. Despite this, Oyster cards issued until January 2010 were still based on the Mifare Classic 1K. This chip has since fallen out of usage in security-conscious environments. Oyster is now based on the Mifare DESFire EV1, which is the object of this study.

The Mifare DESFire EV1 is certified by the German Federal Office for Information Security (Bundesamt für Sicherheit in der Informationstechnik, BSI) under Common Criteria Evaluation Assurance Level (EAL) 4+. It offers a significantly expanded feature set relative to the Mifare Classic. NXP has, for the most part, avoided proprietary protocols (such as CRYPTO-1) in its crypto-coprocessor. It is capable of DES, 3DES and AES-128 encryption, and uses a true random number generator (TRNG) instead of the PRNG found in the Mifare Classic. The TRNG module used to provide security-critical nonces in the DESFire EV1 has not, to our knowledge, been analysed in current literature. We present in here a detailed analysis, reporting consistent and significant biases in the TRNG output from a sample of 100 cards from multiple batches. Our original contributions include an accurate model and a theoretical discussion of the observed bias, and the most comprehensive statistical analysis of an EAL4+ certified smart card's TRNG to date. As the identification of weaknesses in random number generation can aid in the undermining of most crypto-systems, this work may lay the foundations for future attacks against the security of the DESFire EV1. All results have been responsibly disclosed to NXP Semiconductors Ltd. before submission. In subsequent correspondence, they have corroborated our findings but stressed that at present no attacks against this card exist.

The rest of this paper is organised as follows: Section II explores the literature and key technologies. Section III outlines the methods and tools used in this research. Section IV reports on our results and subsequent analysis. Section V concludes the paper, discussing our findings. Section VI discusses future work.

II. Background

As devices that communicate via radio waves, RFID cards are particularly exposed to the risk of their communications being accessed by malicious parties. The fact that the majority of mobile phones are now being equipped with near field communication (NFC, a subtype of RFID) capabilities, including support for card emulation and read functionality, constitutes an additional threat vector.

A. Mifare Classic and Its Security Issues

Mifare is a brand licensed by NXP Semiconductors Ltd. All Mifare cards are based on the ISO 14443 type-A

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standard, communicating at 13.56 MHz. Across all product lines, approximately 10 billion chips have been sold as of 2016 [6]. Introduced in 1994, the Mifare Classic proved extremely popular. Its reliability and ease of implementation led to widespread use in contactless payment solutions. A proprietary cipher, CRYPTO-1, is used to secure the authentication procedure and subsequent communication after successful authentication. The card possesses a PRNG for use in authentication and other cryptographic operations.

Merhi *et al.* reported that the Mifare Classic employs a very weak PRNG [5]. It shows very poor performance in the Dieharder [7], National Institute of Standards and Technology Statistical Test Suite (NIST STS) [8], and ENT [9] batteries of tests. The PRNG was found to be extremely weak, manipulable and vulnerable to pre-emption. The authentication protocol relies on the exchange of nonces, and session keys are derived in part from those. A weak PRNG can and has lead to serious attacks.

The CRYPTO-1 protocol is proprietary to NXP. Its algorithm, undisclosed before its reverse engineering by Nohl *et al.* [10], proved highly vulnerable. The reliance on security through obscurity was shown to be an inferior approach, when in 2008 de Koning Gans *et al.* [3] outlined a practical attack on the Mifare Classic. This attack made use of the weak CRYPTO-1 algorithm to derive private keys after several partial authentications. This allowed unauthorised access to the contents of the card. Further attacks swiftly followed, with Garcia *et al.* [4] demonstrating a reader-based attack on the same year. A compromised RFID card may be cloned, allowing an attacker to duplicate cards for their use, effectively stealing credit from others [11]. It is also possible for individuals to fraudulently top up their e-wallet balances on vulnerable cards. Kasper *et al.* [12] demonstrate examples of such attacks, targeting real-world payment systems.

The Mifare Classic stands as an example of mistakes to avoid when implementing RFID crypto-systems. Security through obscurity offers protection only so long as the obfuscation holds. As demonstrated by Nohl *et al.* [10], once an attacker can look behind the curtain, they may find a weak crypto-system that relied on its obscurity as the primary means of defence. The Mifare Classic has since been patched, but in 2013 patched versions of these cards were again found vulnerable to practical attacks [13].

B. Mifare DESFire EV1

The Mifare DESFire EV1 was introduced in 2006, gaining mass adoption in multiple fare-payment schemes by mid-2010. It is a microprocessor based card that supports multiple applications using a relatively advanced file-system for an RFID card. NPX has implemented open security standards (DES, 3DES and AES-128), with optional authentication for read/write access at the PICC, application and authentication level. The EV1 (2006) is an iteration of the DESFire (2002).

The DESFire EV1 is demonstrably more secure than the Mifare Classic, possessing a TRNG and an AES-128 capable crypto-coprocessor. The use of a TRNG is an improvement from the PRNG used in the Classic, the technical implications of which will be discussed in subsection C. NXP does not disclose the model of TRNG used, which forces us to base our analysis exclusively on its output. Efforts have been made to investigate the power profile of the card, in an attempt to seek side-channel attack vectors [11], [14]. These analyses have found that the DESFire EV1 employs hardware countermeasures that obfuscate power signature fluctuations. These have made the identification of side-channel attacks (SCA) against the EV1 quite challenging. An effective SCA on the DESFire (MF3ICD40) reported by Oswald and Paar [15] has not yet been successfully employed against the DESFire EV1. As a result, no practical attacks against the DESFire EV1 have yet been identified.

C. Randomness and Random Number Generator Testing

The dismantling of the Mifare Classic highlights the importance of robust and sufficiently random nonce generation in RFID cards. Liu and Peng [16] discuss the use of TRNGs to protect against traceability attacks. TRNGs are defined as non-deterministic random number generators (NDRNG), which do not exhibit the entropy-depletion traits of PRNGs.

Methods of extracting random numbers from RFID cards vary slightly between models, but there are some common elements. Verdult *et al.* [17] propose a toolset for RFID protocol analysis, stating that even commercial RFID readers equipped with the Libnfc library are sufficient to begin profiling RFID protocols. Gathering data by this method is slow, as the speed at which a given card may be read represents a hard cap on the rate of data collection.

The PRNGs of the Mifare Classic and Mifare Ultralight-C have been analysed using Dieharder, NIST STS and ENT [5]. The Ultralight-C was found to be significantly more secure than the Classic, but the small dataset (20 MB) may have had an impact on the validity of some results, especially those from Dieharder. Bernstein *et al.* [18] show that just adopting stronger cryptographic algorithms is insufficient to protect cards if other weaknesses are introduced by the supporting hardware. They factor 184 RSA keys out of the 2 million Taiwanese Citizen ID cards using batch-GCD, in part because the RNG shows appalling non-random behaviour for some cards despite being certified CC EAL4+ (as the EV1 under study). A weak RNG is sufficient to compromise an otherwise well-conceived crypto-system.

The testing of random data, once collected, is non-trivial. Dieharder and the NIST STS are two prominent examples in the literature, but ENT [9], TestUo1 [19], and the Avalanche test [20] are other worthy and underused test batteries. When selecting appropriate randomness tests, it is important to consider that they often require large sample sizes to provide reliable observations of a given generator's randomness. Other aspects to consider are the independence of the tests employed and the use cases in which the generator output is likely to be applied.

As the Mifare Classic PRNG proved to be a critical point of failure, the authors deem a thorough analysis of the Mifare DESFire EV1 necessary. Though no practical attacks have been identified yet, the continued security of the

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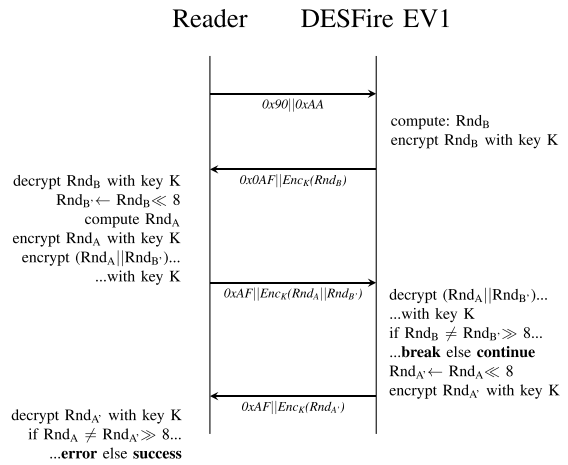

Fig. 1. Mifare DESFire EV1 authentication protocol.

DESFire EV1 relies upon the robustness and reliability of all its security features, including its TRNG.

III. Methodology

A. Data Collection

To test the DESFire EV1 TRNG, the following data was collected:

- 64MB of TRNG output from 3 cards
- 1MB of TRNG output from 100 cards
- 64MB /dev/urandom output

Our sample-set of 100 cards derives from 2 models (DESFire EV1 2k & 4k - identical apart from memory size), 3 retailers and 3 batches. The initial 3 cards (4k) used in our pilot study are all from the same retailer and batch.

A Python script was used to perform the mutual three-pass authentication protocol outlined in Fig. 1. This program was automated with the use of a bash script, allowing repeated authentications to be made in quick succession. This facilitated the reliable collection of random data. The reader's field remained powered between authentications. All authentications were performed at the application level, using an application initialised before the commencement of testing.

By this process, multiple $\mathrm{Enc_K(Rnd_B)}$ (16 bytes) are collected. The recovery of this value from the first response from the card in Fig. 1 was the first stage of our data collection. These numbers are decrypted when recovered, using the card's key. This produces $\mathrm{Rnd_B}$, the random value passed to the reader by the EV1 as part of its mutual authentication protocol. Collected data was stored in binary files 64MB and 1MB in size, with the $\mathrm{Rnd_B}$ data being used in our subsequent tests. $\mathrm{Rnd_A}$ and session key variables are not analysed in the scope of this work.

The data collection process is time-consuming. A minimum of 12 days was required to collect 64MB data from each card. If cards reported errors or were stopped for any reason, experiments were restarted. This was done to ensure uniformity in experimental conditions, and to prevent hardware resets from affecting the outcome. These standards were upheld for the 1MB data collection process, which took approximately 3.2 hours to complete for batches of 5 cards (reading in parallel). For comparative purposes, /dev/urandom was used to generate a 64MB file of random values. This data has been used to highlight where the output from the tested cards deviates significantly from the expected properties of randomness.

The authors have previously reported preliminary findings from 64MB of TRNG data extracted from DESFire EV1 cards using this method [21]. This sample size is large enough to be used in a variety of randomness test batteries, including Dieharder and NIST STS 2.1.2.

TABLE I
Characteristics of Selected EV1 Cards

Model	No.	Auth. Mode	Key	App. ID
DESFire EV1 4k	47	AES-128	zeroes	0x0007c1
DESFire EV1 2k	53	AES-128	zeroes	0x0007c1

B. Experimental Set-Up and Card Configuration

Table I outlines the configuration of the DESFire EV1 cards used in this research. Different memory sizes were tested to determine whether any identified issues are specific to a given model of DESFire EV1.

An ACR122U NFC reader was used to capture random numbers from all cards. The reader was connected, via USB, to an Intel core i7 laptop with 8GB of RAM. One hundred (100) cards from 3 batches were selected. All cards were initialised with the same key (16-byte AES-128 key, initialised to zeroes) and application ID (0x0007c1), with each application containing no data files. Henceforth, the designations Card $1, 2, \ldots, 100$ are used instead of their associated UIDs for brevity.

Cards 1, 2 and 3 were used for the initial study of 64MB of output. Each card was read by the aforementioned Python script, and 64MB of data was collected from each. Cards 1-100 were used to provide 1MB output for further analysis. The 1MB samples collected from cards 1, 2, and 3 are independent of their 64MB samples. These cards were read simultaneously, using a bash script to operate 5 readers in parallel. All readers were ACR122U models from the same supplier, connected to the laptop via a D-LINK powered USB hub.

C. Randomness Testing

All 64MB files derived from DESFire EV1 TRNG outputs, /dev/urandom, and an approximate model of observed TRNG characteristics, were subjected to the following randomness test batteries in our initial study:

- Dieharder [7] applied to 64MB datasets
- NIST STS 2.1.2. [8], [22] applied to 64MB datasets
- ENT [9] applied to 64MB and 1MB datasets
- XOR mask test conducted over all datasets

A matrix providing an overview of the tests performed on each dataset can be seen in Table II. Where the named test is performed on the listed dataset *Yes* is given, where the test is not performed *No* is entered.

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TABLE II
MATRIX OF TESTS APPLIED TO EACH DATA SET

Test	EV1 64MB	EV1 1MB	urandom 64MB
Dieharder	Yes	No	No
NIST STS 2.1.2 2k	Yes	No	No
ENT.	Yes	Yes	Yes
Byte-wise XOR Mask test	Yes	Yes	Yes

All tests are applied over 64MB of data. This amount of data allows for many of the tests to be completed without the issue of multiple rewinds affecting the outcome significantly. 1MB samples are not tested using NIST STS 2.1.2 or Dieharder, as they are of insufficient size to give accurate results. *urandom* is only tested with the ENT and XOR mask tests, as it is used as a control for comparison with the flawed DESFire EV1 data.

The null hypothesis (H_o) for each statistical test is: *The tested data does not exhibit signs of order or structure.* This intuitively leads to the statement that it is expected that tested data will be random. Should the tested data show signs of order or structure, it also follows that it cannot be considered to be random, and H_o should be rejected.

As all of the tests performed share H_o, they all share a common set of definitions for type 1 and type 2 errors. Type 1 errors occur when H_o is true, but rejected. Type 2 errors occur when H_o is false, but not rejected. Below, a context-specific definition of such errors is given.

- Type 1: Test A doesn't suggest ordered/structured data in the test sample, but is considered to possess non-random traits. The sample fails Test A erroneously.
- Type 2: Test A finds evidence of non-randomness in the test sample, but is considered to be sufficiently random, passing Test A erroneously.

Type 1 errors are mitigated by setting an appropriately stringent significance level. The significance level of Dieharder, NIST STS 2.1.2 and ENT tests is the default value: 1% (0.01). This allows an acceptable range of 0.01-1.00 for all p-values resulting from these tests. All tests reporting p-values within this range are considered to accept H_o. All tests reporting p-values of less than 0.01 are considered to reject H_o. Type 2 errors have been mitigated by performing a pilot study on the effect of sample size on the performance of DESFire EV1 TRNG output on the results of statistical tests. It was found that testing less than 500KB resulted in what are likely Type 2 errors for ENT and the XOR mask test. This has informed the minimum sample size of 1MB, which has been selected for this research.

1) Dieharder: Some of these tests had to be modified to prevent excessive rewinds. All sample sizes (p-samples) remained at default values. All tests included in the results that are not in the list below use their default t-sample (number of samples) settings.

- Overlapping Permutations: 125,000 t-samples
- Binary rank 32x32: 4,750 t-samples
- Binary rank 6x8: 25,000 t-samples
- Craps: 20,000 t-samples

The Sums, OPSO, OQSO and DNA tests have been omitted from this experiment, due to their suspect reliability [23].

2) NIST STS 2.1.2: This test battery was run over the 64MB data sets with a stream size of 2,000,000 bits. All STS tests are run over 200 bitstreams so that the minimum representative value of 55 test results would be achieved. The majority of tests report back results out of 200, with Random Excursions and Random Excursions variants tests reporting results out of 118 tests.

EV1 1MB data was too small for reliable use in NIST and Dieharder tests, but ENT and the XOR mask tests are less demanding on minimum sample size. Urandom was not tested using Dieharder, nor NIST STS 2.1.2.

3) ENT: This test battery provides a chi-square test in addition to tests of compressibility, correlation and entropy. It does not possess the wide range of statistical tests possessed by Dieharder and NIST STS 2.1.2, but it provides comprehensive low-level outputs that can be useful. It is also not as commonly used as Dieharder and NIST STS 2.1.2, making it an interesting addition to our low-level test methodology.

A byte-level ENT test was applied over all data-sets. Due to a lack of reports of bias in the DESFire EV1 TRNG, it is assumed that ENT has not been used in this manner before. We validated the output of the ENT chi-square goodness of fit test, by repeating our tests with MATLAB and R implementations of the same byte-level test, deriving identical results.

4) XOR Mask Test: A variable length (8, 16, 32-bit) XOR mask test was used to clarify which values in the data sets contributed most significantly to the observed bias. Like the non-overlapping and overlapping template matching test (NIST STS 2.1.2), this test applies bit-masks of size n over n-bit windows of data. Unlike the tests mentioned above, the window is moved n bit positions after every possible permutation of bits has been XORed against the tested window. Summing all occurrences of a mask that XORs against the tested window - to produce a string of zeroes - allows one to count the occurrence of bit-sequences within a file. The objective is to gain an insight into the over or under-occurrence of given sequences.This test is not considered an original contribution in this work, but its results regarding the DESFire EV1 are.

For the use case defined in this paper, $n = 8$. This creates byte-sized windows, as initial testing with Ent suggests a byte-wise bias that warrants further investigation. Each 8-bit window is XORed with 256 masks, corresponding to all permutations of the n-bit mask length. Masking that resolves to a resulting n-bit run of zeroes increments a counter for that mask. Any other result is not counted. The sum of each mask value may then be visualised, allowing the over or under representation of given values within a set to be observed more clearly. It would not be unfair to state that this is far more a clarification of the results of previous tests than a stand-alone test and the authors strongly encourage the use of this test as a means of refining prior observations.

The 0th mask will always resolve to a value of -0.5, which is merely a consequence of the XOR operation and in no way representative of bias on that mask. As a result, this mask should be ignored unless it deviates significantly from this

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TABLE III
DIEHARDER RESULTS

Test	Card 1	Card 2	Card 3
Birthday Spacings	0.18194520	0.61105583	0.78263630
Overlapping Permutations	0.38044164	0.58693289	0.44201308
32x32 Binary Rank	0.42920693	0.23409500	0.55699838
6x8 Binary Rank	0.31311490	0.32387215	0.66137580
Bitstream	0.97724174	0.18743536	0.59532464
Count the 1's (stream)	0.17108396	0.74984724	0.87214241
Count the 1's (byte)	0.65870385	0.01287807	**0.00020194**
Parking Lot	0.18078043	0.24200626	0.38128677
Minimum Distance (2d sphere)	0.76328000	0.95091635	0.34980807
3d sphere (minimum distance)	0.23871272	0.20826216	0.39340851
Squeeze	0.62598919	0.08843989	0.77057749
Runs	0.99778832	0.62043244	0.90550208
	0.44719093	0.91228597	0.04870531
Craps	0.54077256	0.92769962	0.91803037
	0.57614807	0.94245583	0.95209393

value (which would indicate a problem with the test in most cases).

The XOR mask test can be used to identify the presence and characteristics of bias in purportedly random data. Over or under-representation of one or more values can indicate a bias in the tested sample. This is a useful test to use in conjunction with ENT, which can identify the presence of bias, but is not as useful at isolating it as the XOR mask test.

It is possible to change the size of the window, but for this research, only byte-size windows are reported. Windows of 2, 8 and 16 bytes in length were also tested but returned no significant results and are not reported on in this manuscript. From this point the term mask is used interchangeably to discuss both the mask values and the size of the testing window, as both window and mask must be of equal size n.

5) Responsible Disclosure: All findings pertaining to the Mifare DESFire EV1 have been responsibly disclosed to NXP Semiconductors Ltd. A response acknowledging our correspondence and confirming the observed bias has been received.

IV. RESULTS & ANALYSIS

A. Dieharder Results

The results of the Diehard tests can be found in Table III. All three cards perform well, though card 3 exhibits a marginally weak result (shown in **bold**). There is no statistically significant evidence of weakness in the cards, although the third card showed some borderline results. Dieharder did not detect any significantly non-random behaviour in the DESFire EV1 TRNG over 64MB of data.

B. NIST STS 2.1.2 Results

Table IV, shows the results of NIST STS 2.1.2 performed over the 64MB datasets.

As with Dieharder, there is no statistically significant evidence of non-randomness in the three cards that were tested. Cards 1 and 3 did show some borderline results, but not to a degree that would suggest that the TRNG's were deficient.

Considering the previously discussed issues of TRNGs and their interaction with single-run randomness testing, it is likely

TABLE IV
NIST STS 2.1.2 RESULTS

Test	Card 1	Card 2	Card 3
Frequency	198/200	200/200	197/200
Block Frequency	196/200	199/200	194/200
Cumulative Sums	2/2	2/2	2/2
	(>193/200)	(>193/200)	(>193/200)
Runs	**191/200**	197/200	**192/200**
	(p≈<0.001)	(p≈0.719747)	**(p≈0.007160)**
Longest Run	196/200	198/200	198/200
Rank	198/200	199/200	197/200
FFT	197/200	199/200	198/200
Non-Overlapping Template	**147/148**	148/148	148/148
	(>193/200)	(>193/200)	(>193/200)
Overlapping Template	198/200	198/200	198/200
Universal	198/200	198/200	198/200
Approximate Entropy	197/200	198/200	196/200
Random Excursions	8/8	8/8	8/8
	(>113/118)	(>114/118)	(>113/118)
Random Excursions Variant	18/18	18/18	18/18
	(>113/118)	(>114/118)	(>113/118)
Serial	2/2	2/2	2/2
	(>193/200)	(>193/200)	(>193/200)
Linear Complexity	199/200	197/200	199/200

TABLE V
MIFARE DESFIRE EV1 ENT RESULTS FOR 64MB OF TRNG OUTPUT

	Card 1	Card 2	Card 3	Optimal
Entropy	7.999969	7.999989	7.999972	8
Optimal Compress.	0	0	0	0
Chi-Square	**2709.10**	**973.07**	**2470.32**	256
Arith. Mean	127.492921	127.500582	127.5006	127.5
Monte Carlo π est.	3.14167	3.142019	3.141909	3.14159
S. Correlation	0.000008	0.000045	0.000093	0.0

that marginally failed results may merely be indicative of normal TRNG operation. Multiple rounds of testing would be required to identify a consistent failure. Due to the amount of time required to collect 64MB samples, this was not possible within the scope of these experiments.To further explore these results, ENT was used to test the TRNG data for each of the cards.

C. ENT Results

All three cards perform well on the majority of ENT tests. Table V shows that the Entropy, Optimal Compression, Arithmetic Mean, Monte Carlo and Serial Correlation tests all report results within optimal parameters. The results of the chi-square test, however, tell a very different story.

All three cards perform exceptionally poorly in the chi-square test. These are not small deviations from the optimal value of 256, as all three cards are at least at 3.8 times the optimal value, with card 1 proving especially weak at 10.5 times the expected average.

It is permissible for the values reported by ENT to be higher or lower, but significant deviation from the average hints towards a systemic bias in the tested data. The optimal chi-square score will vary with the size of the test window. All tests are conducted on 8-bit windows in this work.

In conjunction with this chi-square score is a p-value, which should be between 0.1 and 0.9, but is tolerable within

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TABLE VI
MIFARE DESFIRE EV1 ENT RESULTS FOR 1MB OF TRNG OUTPUT

	Card 1	Card 2	Card 3	Optimal
Entropy	7.999780	7.999820	7.999786	8
Optimal Compress.	0	0	0	0
Chi-Square	**305.47**	249	**297.03**	256
Arith. Mean	127.6015	127.5626	127.5082	127.5
Monte Carlo π est.	3.13620558	3.140892564	3.140388562	3.14159
S. Correlation	-0.000068	0.001339	-0.001751	0.0

the bounds 0.01 and 0.99 for generators used for non-cryptographic purposes. All of the cards that have been tested reported p-values below 0.01 for this test, indicating that the data should not be considered random.

The chi-square results are sufficiently high and consistent to force us to conclude that there is enough evidence of strong non-random behaviour in this TRNG. Despite passing all tests in Dieharder and STS 2.1.2, all cards fail the ENT chi-square goodness of fit (GoF) test. The severity of the results, ranging from 973.07 to 2709.1 (expected value is 256), makes it clear that there is some form of systemic bias occurring at the byte level. Although card 2 performs better on the chi-square than the other cards, there is variability between cards 1 and 3 also. This suggests that the severity of the observed bias may vary between cards. This is a potentially distinguishing feature.

The need to collect data from a large number of cards (100) required the identification of the minimum working sample of data that would exhibit this bias. Reduction of the sample size of the three initial cards, through 32MB, 16MB, 8MB, 1MB and 0.5MB found that 1MB samples retained evidence of the poor behaviour of the EV1 TRNG. This expedited the collection of a more representative sample from 100 cards.

Table VI shows the 1MB results for the initial cards that we tested. Cards 1 and 3, yet again, show that they perform poorly compared with card 2. Card 2, in fact, passes the chi-square test within acceptable parameters. The identification of the bias at this smaller data size allowed the collection of 1MB samples from 100 EV1 cards in a timely manner.

Fig. 2 shows the ENT chi-square goodness of fit results for 100 DESFire EV1 cards. Each point represents the scores (*y-axis*) of specific cards as denoted by the *x-axis*. These results show that 78% of the cards tested fail the chi-square test, reporting values over 300. The mean score, represented by the unmarked black line, is 314.17, which is far beyond the expected 256 an unbiased RNG should show. The upper-bound is extremely high, with 5% of tested cards showing scores over 400. Only 9% of cards demonstrated results equal to or lower than the expected score (denoted by the lowest broken line). This demonstrates that not only does the poor performance hold throughout a larger sample of DESFire EV1 cards, it is far worse in some cases than our initial study suggested. In conjunction with the previously discussed chi-square results, this demonstrates a clear and consistent issue with the DESFire EV1 cards that have been tested.

D. Initial Analysis of Bias

Further analysis was undertaken to determine the nature of the observed bias. Fig. 3 shows the mean bias for the

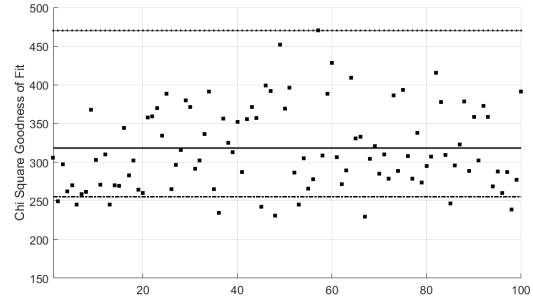

Fig. 2. Chi-Square Results for 1MB of data from 100 DESFire EV1 cards.

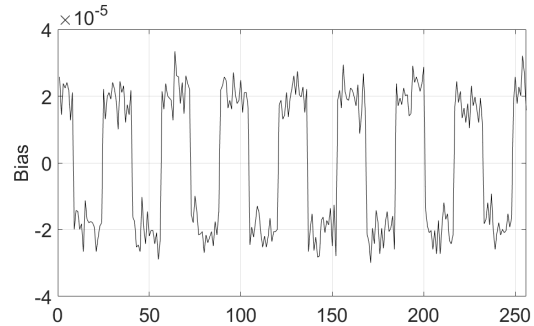

Fig. 3. Mean bias of 64MB of data from 3 DESFire EV1 cards.

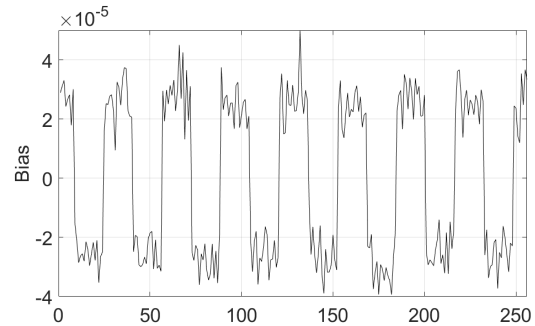

Fig. 4. Mean bias of 1MB of data from 100 DESFire EV1 cards.

64MB samples generated by cards 1, 2 and 3 during initial testing. The *x-axis* represents the decimal integer value of the bytes tested. As these analyses are conducted over single bytes, this ranges from 0-255. A regular sinusoidal pattern can be seen in the bias of the TRNG output. The number of more frequently occurring values is equal to that of the less frequent values (128 values in each case). The peaks and troughs of the line display some noisy behaviour but do not detract from the general pattern. This pattern suggests that the underlying generator has a significant bias. The maximum positive bias for the mean 64MB sample is 3.3426×10^{-5}, while the maximum negative bias is -2.9908×10^{-5}.

Cards 1, 2 and 3 showed varying magnitudes in their biases. Card 3 exhibited the most substantial bias, with card 2 showing the least. These observations correlate with the chi-square test results: larger scores on that test translate into more substantial observable biases.

Fig. 4 reports the mean bias observed over 100 EV1 cards, taken from 1MB of data. The bias is slightly larger than that

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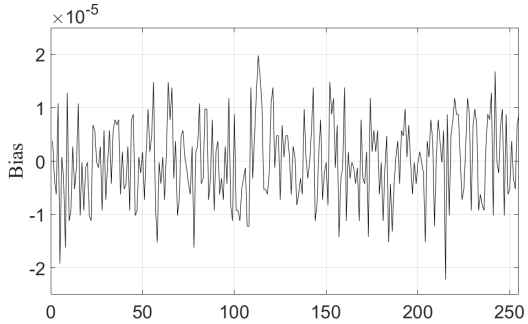

Fig. 5. Bias values for 64MB of /dev/urandom data.

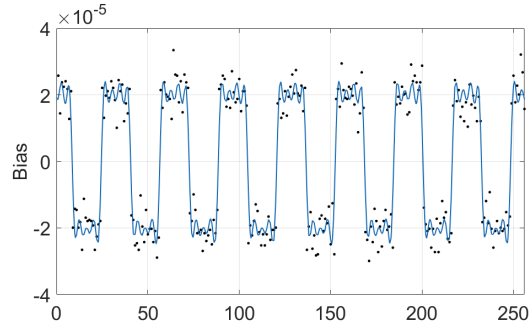

Fig. 6. Fourier approximation for 64MB of data from 3 DESFire EV1 cards.

seen in Fig. 3. This is due to the poor performance from many cards in this data set, relative to the 3 initial cards tested. The same regular pattern observed in Fig. 3 can be seen. Similar extremes are observed in the bias (4.9908×10^{-5} and -3.9312×10^{-5}). The distribution of values between peaks and troughs is exactly even (128 each way). Individual cards showed far noisier output at 1MB than at 64MB, which is expected. However, 78% of cards displayed an easily identifiable bias. All cards displayed bias when subject to this level of scrutiny, regardless of their chi-square scores.

For comparison, Fig. 5 shows the bias of a 64 MB sample taken from the Ubuntu 16.04 implementation of /dev/urandom. Its chi-square value is 250.15 with a p-value of 0.2513. There is no immediately discernible pattern to the output of this PRNG. The maximum bias values for this sample are 1.98×10^{-5} and -2.22×10^{-5}. Most notable is the absence of regular sinusoidal patterns in the distribution of bias. This outlines the deviation of the EV1 TRNG from expected results. This is sufficient to conclude the TRNG is not random enough to adhere to the requirements of BSI AIS 31 [24] at the consumer level.

E. Analysis of the Trend

To analyse the observed trend, a Fourier approximation of the biases of each card's data set was computed:

$$P(x) \approx a_0 + a_1 \cdot cos(x \cdot w) + b_1 \cdot sin(x \cdot w) + (\ldots) + a_8 \cdot cos(8 \cdot x \cdot w) + b_8 \cdot sin(8 \cdot x \cdot w) \quad (1)$$

Equation 1 shows the Fourier approximation, where x may take any value between 0 and 255. Plotting the fitting curve helps identify the pattern of the bias seen in the previous figures.

Fig. 6 shows the resulting scatter plot and trend for the three 64MB EV1 samples. A distinctive sinusoidal trend can be seen, with a period of $w = 31.9918$. This translates to a series of 8 oscillations, representing sequences of approximately 32 values.

Interestingly, a period of 32 suggests the under-occurrence of a specific bit in each byte: bit 5 to be precise (assuming little-endianess). Mask testing and the development of a model validates and improves on this initial observation, in subsequent subsections of this work. The distribution of data points shows a tendency for values to deviate from the expected

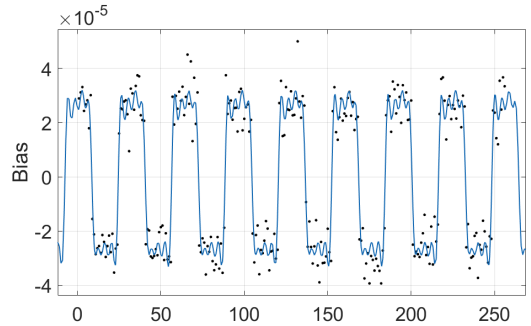

Fig. 7. Fourier approximation for 1MB of data from 100 DESFire EV1 cards.

distribution. 239 values out of 256 possess bias greater than $\pm 0.5 \times 10^{-5}$. This is a significant deviation from the expected values in a random dataset. More than half of the possible values in the mean of the 64MB datasets are biased by more than 50% of the observed extremes. The mean of 100 1MB EV1 samples demonstrates a very similar trend, as can be observed in Fig. 7. As with the previous observations, the bias is slightly larger, than that shown in Fig. 6. The previously discussed rationale for this phenomenon stands.

A period of 31.87 can be derived from the data. This is very close to that shown in Fig. 6, iterating over 32-byte values between peaks. This reinforces the notion of a systemic problem, suggesting a bias towards bit 5 in each byte being 0 more often than expected. The distribution of values on the y-axis is very similar to that observed in Fig. 6. The consistent deviation from the normal previously observed for the 64MB samples remains here, with a larger magnitude. The empty channel around the normal is also witnessed here, with similar upper and lower bounds. This is due to the proportion of well-behaved cards being very low, comprising less than 22% of the tested cards.

Fig. 8 shows the Fourier approximation for a 64 MB data set captured from /dev/urandom. The distribution of values over the y-axis is visibly different to that of any of the EV1 data sets.

As previously mentioned when discussing the plots of the bias for the Mifare DESFire EV1's and urandom, the y-axis channel in which the values reside is at least 50% smaller than any observed in the TRNG data sets. The trend seen in the other three figures has completely disappeared, due to

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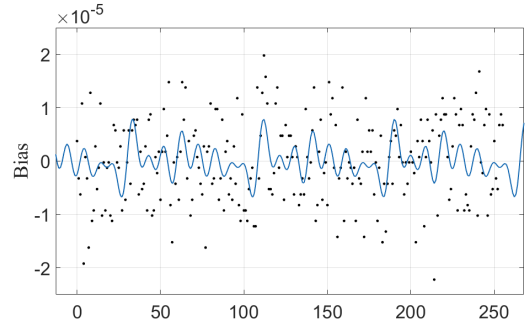

Fig. 8. Fourier approximation for 64 MB of urandom data.

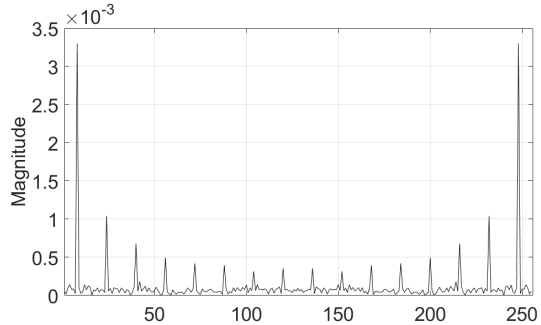

Fig. 9. Fourier approximation for FFT of 64MB fitting curve.

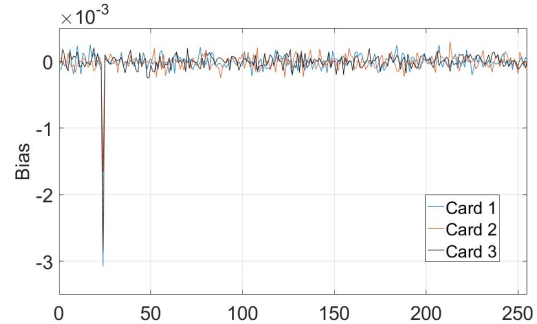

Fig. 10. Composite mask results for 64MB of data from 3 DESFire EV1 cards.

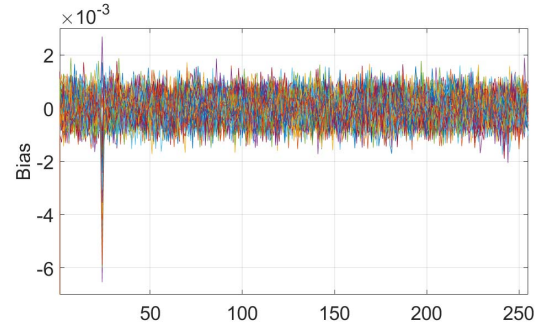

Fig. 11. Composite mask results for 1MB of data from 100 cards.

the effects of a more even distribution throughout the range of bias. Compared with the TRNG output, urandom is very evenly distributed across the y-axis, with only 52 values lying either side of a $\pm 1 \times 10^{-5}$ y-axis exclusion channel. This drops sharply to 10, for a $\pm 1.5 \times 10^{-5}$ channel. Only 1 value can be found beyond a $\pm 2 \times 10^{-5}$ channel. This serves to highlight the demonstrable pattern of non-randomness in the Mifare DESFire EV1 data. Fig. 9 depicts the Fast Fourier Transform (FFT) of the fitting curve shown in Fig. 6.

F. XOR Mask Test Results

Masking tests were performed at mask sizes of 1 byte, 2 bytes, 8 bytes and 16 bytes to determine which values exhibited the greatest bias. Only the single byte tests showed any anomalous behaviour, and so the other results are omitted from this publication for the sake of brevity.

Fig. 10 shows the masking test results for the 64MB data set. This is a composite graph, showing the biases for each mask (represented on the x-axis by integer values from 0 to 255).

All three cards show significantly fewer occurrences of mask 00011000 than expected. The most extreme of the cards (3) shows a bias of -3.07×10^{-3}. Card 2 is the least biased but still demonstrates a significant problem, with mask 00011000 showing a bias of -1.65×10^{-3}. All other values exhibit biases in the order of 10^{-4} at most. For cards 1 and 3, the bias on mask 00011000 is over an order of magnitude greater than the next largest bias. Furthermore, there does not appear to be any correlation between the biases seen in other masks when comparing cards.

Fig. 11 shows the masking test results for all 100 1MB samples taken from the DESFire EV1's. The bias of mask 00011000 is twice as great as that seen in Fig. 10, for the worst performing cards. 78% of tested cards demonstrated a bias on mask 00011000 in excess of -2×10^{-3}. A single card from this set showed a bias of 2.79×10^{-3} for this mask.

This is an anomalous result, but subsequent testing of the card demonstrated that this bias was observed in two more 1MB samples. At this time, we do not have an explanation for this inversion. It is hypothesised that a manufacturing flaw, possibly in the implementation of a whitening function, has contributed to this anomaly, and possibly the bias seen in other cards. Card 49, the worst performing card, demonstrates an extreme bias of -6.09×10^{-3}, and the bottom 5% of cards exhibited chi-square test scores of over 400.

A significant under-occurrence of mask 00011000 characterises cards that perform this poorly. Only 5% of cards demonstrate bias this extreme, but this becomes statistically significant when one considers that the DESFire EV1 is distributed in figures exceeding 8 million units a year by just one vendor (TfL.) in the UK. Should these results hold over the whole DESFire EV1 family, this becomes a not inconsiderable number of cards with poorly performing TRNGs. TRNG output is critical to the function of authentication protocols for access, read and write functionality. Should practical attacks exploiting this bias be developed, this could be a very serious problem.

G. Modelling

Through observation of the Fourier approximations for the DESFire EV1, one may infer that the 5th bit of each byte (b_5), plays a significant role in the observed bias. Masking tests confirmed that mask 00011000, linking b_4 and b_5, is

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TABLE VII
ENT RESULTS FOR 64MB OF MODEL OUTPUT

	Model	Optimal
Entropy	7.999989	8
Optimal Compress.	0	0
Chi-Square	**934.486344**	255
Arith. Mean	127.500748	127.5
Monte Carlo π est.	3.141804	3.14159
S. Correlation	0.000013	0.0

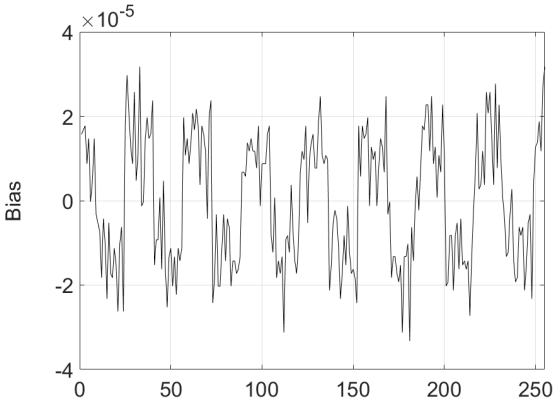

Fig. 12. Bias analysis of 64MB of model data.

maximally biased. This leads to the hypothesis that b_4 and b_5 are in some way correlated, as the results show that the distribution of $b_4 \wedge b_5$ is exceptionally strange.

A model has been created which uses the Python random library to generate 8 bits in a loop, representing the generation of a byte. In each loop, there is a probability p that b_5 will take the same value that b_4, biasing the otherwise random output. For this work, a probability of $p = 0.0031$ was tested as an initial approximation to this pseudorandom data.

Table VII shows the ENT results for the data set captured from the model. As with previous results, all tests except for the chi-square provide insufficient evidence of non-randomness. The chi-square for the model is close to that of card 2, despite using values derived from card 1 to calibrate the model. The means by which the bias is introduced to the model (random allocation of biased bits) is most likely responsible. This can lead to fewer biased bytes than expected in this particular case.

A plot of the model bias, shown in Fig. 12 (b), demonstrates a relatively similar pattern to that observed in the Mifare DESFire EV1 TRNG datasets (Fig. 12 (a)). The model exhibits characteristics closer to that of card 2 than to cards 1 or 3. When compared with the mean bias of all three EV1 cards, it shows more noise on transitions between peak and troughs along the trend.

The model bias ranges from 3.8750×10^{-5} to -3.3250×10^{-5}. This is analogous to the range of bias observed for card 1 but slightly higher than the mean (3.3426×10^{-5} to -2.9908×10^{-5}). The model generated 126 values that were more likely than the other 130, approximately matching the characteristics of the mean EV1 TRNG data (128 both ways).

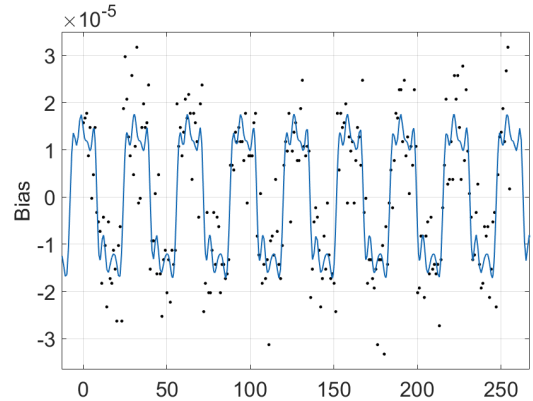

Fig. 13. Fourier analysis of 64MB of model data.

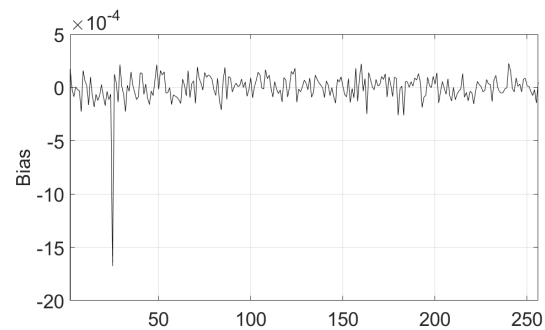

Fig. 14. Mask test results for 64MB of model data.

When analysing sufficient data, a roughly even distribution of values either side of the normal is expected in a random system. It is the distribution between the upper and lower bounds that can call the randomness of the observed system into question.

Fig. 13 shows that the model (b) has a similar pattern to that of the mean bias of the EV1 TRNG (a). The magnitude of that bias is particularly reminiscent of card 2, with which the model shares a similar chi-square result. Similar to the better performing EV1 cards, the model shows a more even distribution of values across the y-axis but falls far short of urandom in this regard. Out of 256 values, 202 can be found either side of a $\pm 1 \times 10^{-5}$ exclusion channel. This is less than the mean, which has 253 values outside this channel. Increasing the size of this channel to $\pm 2 \times 10^{-5}$ reduces this to 90 values. The mean is reduced to 132 values under these conditions. This is noteworthy, as although the values are certainly more distributed over the y-axis than in the DESFire EV1 results, the larger extremes and more numerous values at these extremes cause the model to exhibit behaviour similar to the majority of the tested cards. The more even distribution across the y-axis in the model implies that the behaviour displayed by the TRNG may not be wholly probabilistic.

Fig. 14 shows the mask test results for the model. It displays the same bias on mask 00011000 that has been seen in the DESFire EV1 TRNG results. The bias on mask 00011000 easily exceeds 10^{-5}, reaching -1.67×10^{-5}. There are no noticeably excessive biases on other values, though the model

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TABLE VIII
COEFFICIENT OF DETERMINATION (R^2)

	R^2	Adjusted R^2
Card 1	0.7981	0.7836
Card 2	0.7288	0.7094
Card 3	0.8121	0.7987
mean (64MB)	0.8804	0.8612
mean (1MB)	0.8703	0.8591
urandom	0.0479	-0.0201

shows a more normalised distribution throughout all masks except 00011000. This is expected, as the base RNG for this model is pseudo-random, instead of non-deterministic.

Table VIII shows the coefficient of determination for each of the data sets used in this paper. Card 1 and 3 show R^2 values greater than 0.79, indicating a more than 79% match with the pattern. Though not as high, the R^2 for card 2 demonstrates a >72% match with the model. The mean of the 64MB and 1MB data sets reinforces the notion of this as a preliminary model of the *general* bias of the TRNG in question. Both sets of DESFire EV1 TRNG output demonstrate a correlation coefficient of more than 0.87, indicating an 87% match with the model output. These results indicate a significant deviation from randomness, made more apparent when compared with urandom, which returns an R^2 of only 0.04. We can conclude that the model is a good initial representation of the bias identified in the Mifare DESFire EV1 TRNG, but further research could increase its accuracy by identifying other minor contributing factors.

V. CONCLUSION

An in-depth analysis of the Mifare DESFire EV1 TRNG has been presented. ENT results indicate significant issues with the randomness of 78% of DESFire EV1's tested. Clear, consistent and significant biases have been identified and traced to a specific anomaly in the TRNG output. These findings have been responsibly disclosed to NXP, who have responded, confirming that they were able to reproduce the bias independently.

Fourier approximations to the bias produce a distinctive, sinusoidal, pattern in the TRNG data. A period of approximately 32 can be observed for each card, indicating a significant bias in the TRNG output. This shows that the TRNG output of the DESFire EV1 is decidedly non-random, despite using a non-deterministic random number generator. This leads to questions regarding the certification of such a device at EAL4+, which requires compliance with AIS 31 RNG guidelines. Analysis of the masking test results shows that the DESFire EV1 TRNG bias lies mostly on the mask 00011000. This confirms that the 4th and 5th bits of a significant proportion (approximately 0.38%) of bytes are related. No other correlations have been identified in the bias at this time.

A model based on these findings has been developed. The coefficient of determination for this shows that it can explain approximately 87% of the observed patterns for the mean of both 64MB and 1MB data sets. This suggests that the relationship between bits 4 and 5 of a proportion of bytes is the primary source of the problem. Identifying secondary contributing characteristics is an item of ongoing work. No practical attacks have been identified in this paper. The authors believe, however, that continued analysis of the TRNG, in conjunction with other vulnerabilities, may pave the way to future practical attacks on the Mifare DESFire EV1.

VI. FUTURE WORK

Implementing a non-deterministic source of randomness for the model will be a significant step forward in identifying the faults in this particular TRNG implementation. By applying different whitening functions over a variety of TRNG modules, it may be possible to identify the specific combination that leads to the issues reported in this work.

Testing other Mifare DESFire products to determine whether they share the reported weaknesses, is another item of interest. Mifare Plus X, Plus EV1 and DESFire EV2 are all described as using a TRNG, which may indicate that they are using the same module. If this is the case, any weaknesses found in the EV1 may be relevant for these products. This may also be true for different brands using similar technology (particularly where authentication protocols are provided on-chip), such as Felica.

Analysis of the hardware basis for these errors would be beneficial. Mapping the model to design choices or manufacturing error is one area of further inquiry. A full analysis of the different biases between cards, based on their date of manufacture, batch and other identifying characteristics would also help identify further characteristics of the bias. Mapping such characteristics to the physical placement of each chip on the original wafer from which it was selected, may go some way to highlighting a systemic issue with the manufacturing process.

Building on these findings, the authors will put forward a critique of the EAL4+ certification applied to the Mifare DESFire EV1. The aim will be to develop new guidelines for the implementation and testing of non-deterministic RNGs in resource-constrained contexts.

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